

Days 4–6 Deliverables

Dataset Design & Quality • Pipeline & Experiment Planning • Pre-Dataset Readiness & Mentor Review

Battery Health & RUL — Deep Learning Internship

TIH IIT Guwahati

September 2026

Dataset Design & Quality Expectations

Dataset Schema — Key Fields

Field	Type	Quality Note
cell_id	string	Consistent across every cycle
cycle	int	Monotonic per cell
voltage / current	float	Out-of-range = sensor fault
capacity	Ah	Basis for SOH computation
temperature	°C	Frequently missing
target: RUL/SOH	int/%	Per-cell, not fleet-wide

Train / Val / Test Split Must Preserve

- ▶ Group split by cell_id — no cell spans two sets
- ▶ Chronological order kept within each cell
- ▶ Scalers/encoders fit on train cells only

Dataset Risk Checklist

1. **Missing-Data Risk** — Sensor dropouts mid-cycle; temperature channel often absent.
2. **Annotation Risk** — EOL/target-label definition inconsistent across cells.
3. **Imbalance Risk** — Few near-EOL cycles vs. many healthy cycles.
4. **Temporal/Leakage Risk** — Future-cycle features or fleet-wide scaling before split.

Pipeline & Experiment Planning

Reproducible Pipeline



Baseline (ML)

- ▶ Linear / Ridge Regression
- ▶ Random Forest Regressor
- ▶ Gradient Boosting (XGBoost)

Advanced (DL)

- ▶ LSTM / GRU
- ▶ 1D-CNN on V/I curves
- ▶ Transformer-based RUL

Eval Metrics

- ▶ MAE, RMSE, MAPE, R^2
- ▶ Train / inference time
- ▶ Model size

Project-specific experiment

Compare feature-based ML vs. sequence-based DL for **both** RUL and SOH on the provided dataset, under the same leakage-safe split.

Pre-Dataset Readiness + Mentor Review

Package to Complete

- ▶ Literature review summary
- ▶ Methodology V1 document
- ▶ Data-readiness checklist (Day 4)
- ▶ Preprocessing/modelling/eval plan

To Confirm with Mentor

- ▶ Split strategy (group by cell_id)
- ▶ Evaluation metrics & priority order
- ▶ Compute / time-budget constraints
- ▶ EOL / target-label definition

Post-review action

Incorporate mentor corrections → re-issue Methodology V1 → V1.1, with an updated experiment plan reflecting the confirmed split, metrics and constraints.

Days 4–6 Deliverables at a Glance

Day 4 — Dataset Design & Quality
Schema note • Risk checklist

Day 5 — Pipeline & Experiment Planning
Flowchart • Plan • Log • Folder

Day 6 — Readiness + Mentor Review
Methodology V1 • Updated plan • Corrections